



UNIVERSITI TUNKU ABDUL RAHMAN

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

UCCD3084 GRAPHICS PROGRAMMING FOR XR

Project Title:

Fire Extinguisher Training VR Simulation in a Factory

Factory Type:

Mannequin Factory

Group Number:

16 (NerveGearVR)

Group Leader:

Hew Yu Jing

Submission Date:

8/9/2025

Name	Student ID	Practical Group	Programme
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Project Scope & Environment Design

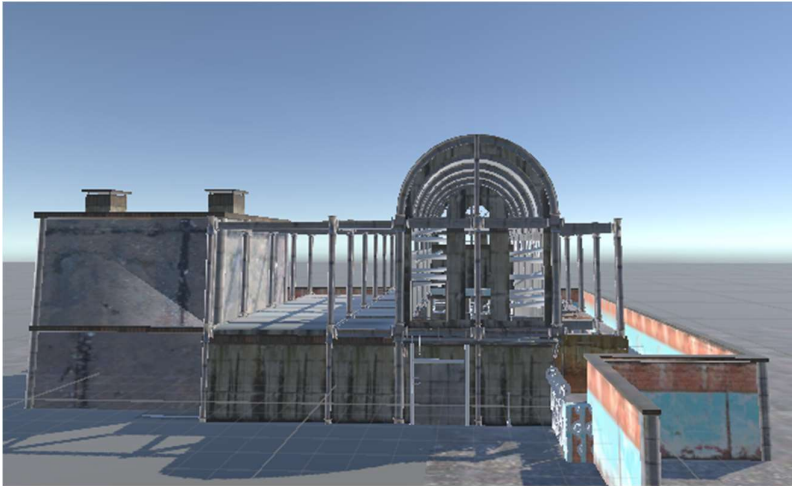
The project scope covers the techniques of using PASS (Pull Aim Squeeze and Sweep). The user is required to pull the pin by using another hand to grab the pin. Then the user is required to aim at the fire, squeeze and sweep the fire extinguisher.

We use a Mannequin Factory for our assignment. It is a rundown factory storing mannequins to give it a creepier and more dangerous vibe. It has an array of fire hazards such as faulty electricity systems and gas leaks. In this assignment, the fire is caused by a gas leak from the pipes and overloaded electrical systems.

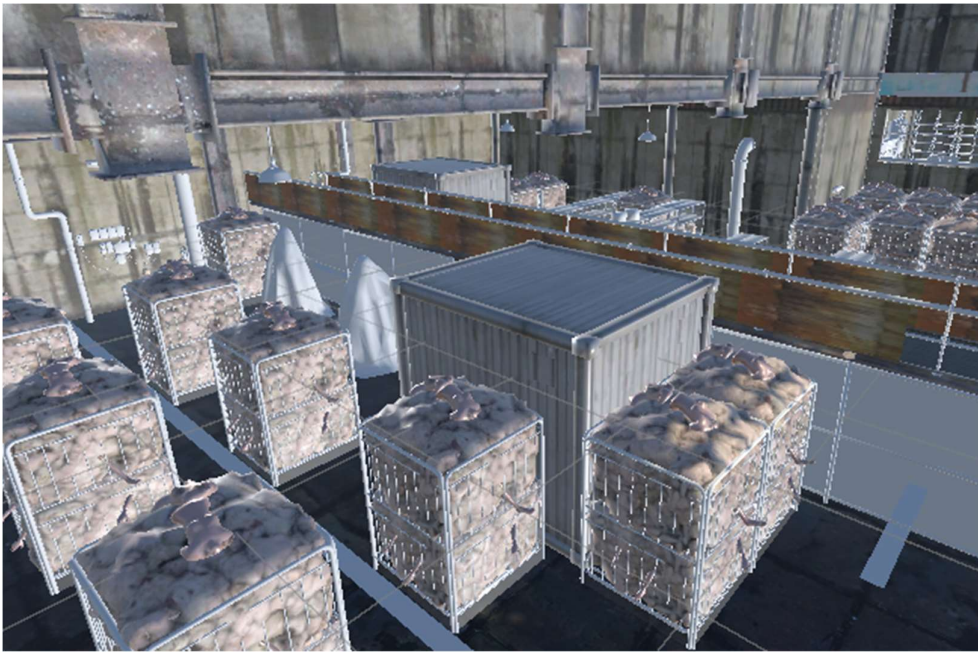
The fire extinguisher is placed outside the factory to prevent it from explosion. It is placed right at the entrance of the factory to ensure easy access.

We constructed the trainee as an action-based XR rig (XR Origin + Input System) using the Ren avatar to ground it, a camera mounted on the device (at the height of approximately 1.7 m), and a gaze-congruent movement. The Character Controller is configured to train at a pace; height = 1.8, radius = 0.22, slope = 60, step offset = 0.3, small skin width, with Continuous Move (gravity Try Attempting Move) and Continuous Turn; it receives the XRI default action maps via an Input Action Manager. The XR Device Simulator is configured in-Editor with Game View of All Device Input. The extinguisher and pin are XR Grab Interactables (rigidbodies, sized colliders, attach points) and the PASS logic is an XRI action listener (no legacy UnityEngine.Input).

The factory was physicalised, giving it a visual only, where boxes were made in the form of colliders, single convex meshes were used where necessary, an invisible ramp collider (stairs) made, the world on Default layers, the grabbables on interaction layers, the rig layer overrides the environment, the rig layer does not include the UI. Main fixes: pink materials: create/assign URP assets and reconvert input exceptions: replace legacy Input reads with actions stair snagging: ramp + small skin width/min move distance mouse not captured: use Device Simulator UI and lock all input to Game View.



Exterior of the building.

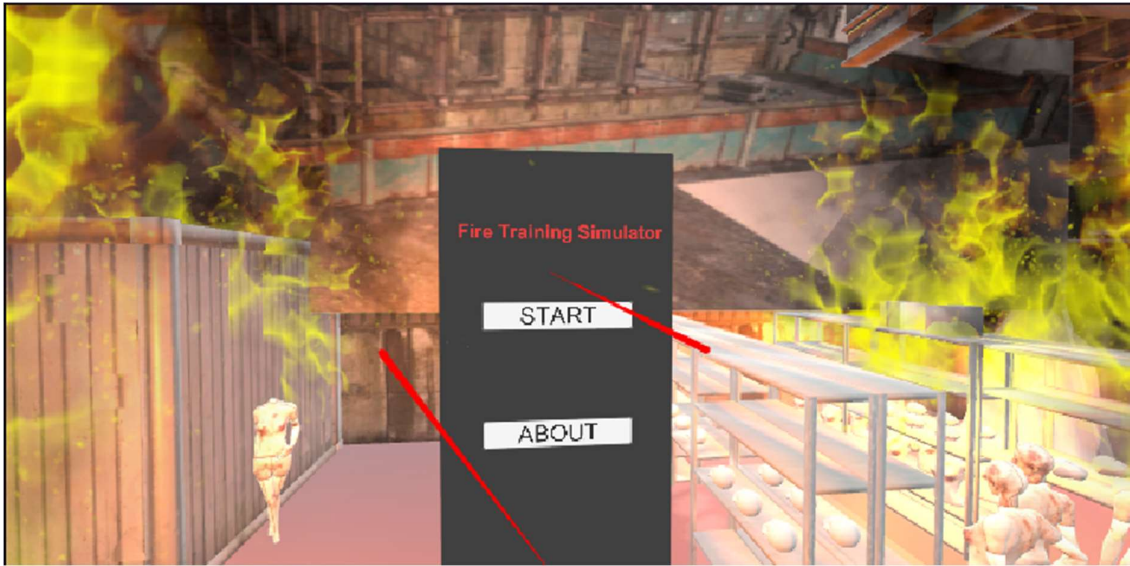


Interior of the factory. Boxes contains a chunk of mannequin body parts.

Simulation Flow

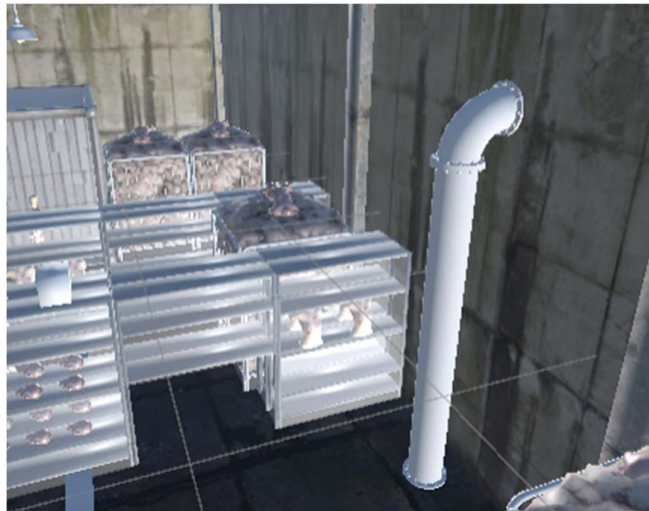
Fire Extinguisher VR Simulation Storyboard

1) Main menu



Before the game started, a game main menu is shown is displayed, two options; Start and About. After pressing start, the player will be spawn outside the realistic mannequinn factory.

2) Fire hazard detected



A fire hazard is detected- Gas leakage from the pipes.

- 3) Get fire extinguisher outside the factory



User navigates using teleportation or joystick locomotion. Highlighted outline or glow effect guides user toward it. User reaches out and grabs the extinguisher using VR hand controls.

- 4) Pull the pin



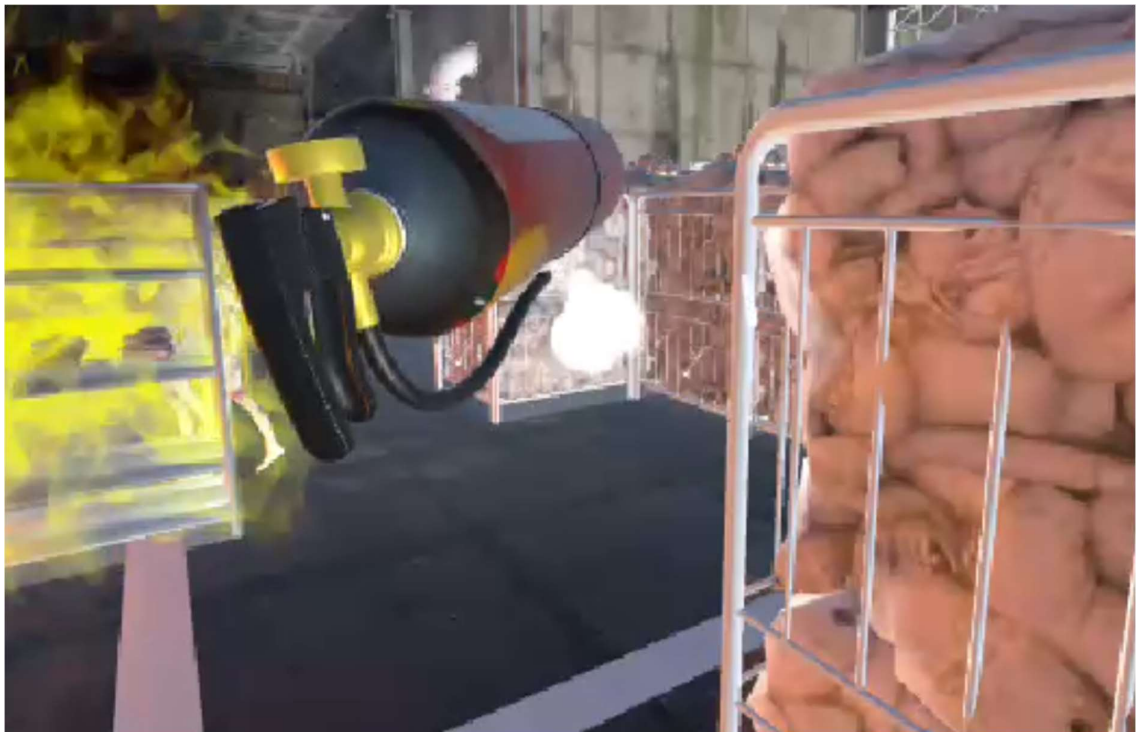
User must locate and pull the safety pin from the extinguisher. A tactile animation shows the pin being removed.

5) Aim the nozzle



User positions the nozzle toward the base of the fire. A laser guide or reticle helps align the aim.

6) Squeeze the handle



User presses the VR trigger to squeeze the handle. CO₂ discharge animation plays: white vapor, hissing sound.

7) Sweep side by side



User moves the extinguisher nozzle in a horizontal sweeping motion. Fire responds dynamically—shrinking with each pass.

Team Contribution Summary

	NAME	CONTRIBUTION
1	Hew Yu Jing	PASS mechanism Factory assets
2	Teng Shiaw Rong	Avatar Character Control Solid Colliders for Factory Walls
3	Ooi Yi Thong	Fire VFX Fire behaviour Open Door Function
4	Seng Qi Zhi	Main Menu Route navigation